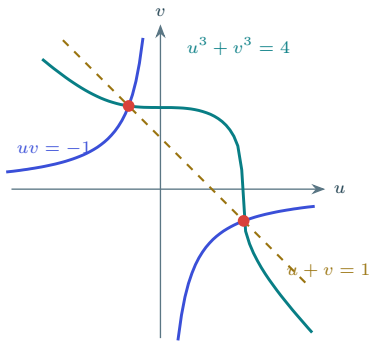
ALGEBRA ON ONE PAGE

Cardano's cubic as a hyperbola

Writing $t = u + v$ with $uv = -p/3$ turns the depressed cubic into the line and hyperbola of the quadratic, followed by a cube root.

1 / THE RULE AND THE PICTURE

The root of $t^3 + pt + q = 0$ is $t = u + v$ at a point on $uv = -\frac{p}{3}$ and on $u^3 + v^3 = -q$



Putting $t = u + v$ gives $t^3 = u^3 + v^3 + 3uv\,t$, that is $t^3 - 3uv\,t - (u^3 + v^3) = 0$, which is the reduced cubic when

$$uv = -\frac{p}{3}, \qquad u^3 + v^3 = -q$$

For $t^3 + 3t - 4 = 0$ the hyperbola is $uv = -1$ and the curve is $u^3 + v^3 = 4$, and they meet at $(\varphi, -\frac{1}{\varphi})$ and its mirror image, both on the line $u + v = 1$, so the root is $t = \varphi - \frac{1}{\varphi} = 1$. The two real points are mirror images in $u = v$, and the four other crossings of a conic with a cubic are complex.

2 / THE TWO LAYERS / A QUADRATIC, THEN A CUBE ROOT

Cubing sends the hyperbola $uv = m$ to $z_1 z_2 = m^3$, so u^3 and v^3 solve the line and hyperbola quadratic

step	rule	$t^3 + 3t - 4 = 0$	$t^3 - 3t - 1 = 0$
1. constants	$m = -\frac{p}{3},\ c = -q$	$m = -1,\ c = 4$	$m = 1,\ c = 1$
2. resolvent	$z^2 - cz + m^3 = 0$	$z^2 - 4z - 1 = 0$	$z^2 - z + 1 = 0$
3. roots	$z_{1,2} = \frac{c}{2} \pm \sqrt{\frac{c^2}{4} - m^3}$	$2 \pm \sqrt{5}$	$e^{\pm i\pi/3}$
4. cube roots	$u = \sqrt[3]{z_1},\ v = m/u$	$u = \varphi,\ v = -\frac{1}{\varphi}$	$u = e^{i\pi/9},\ v = e^{-i\pi/9}$
5. the root	$t = u + v$	1	$2\cos 20^\circ = 1.879$

Steps 2 and 3 are the quadratic sheet with $s = c$ and $P = m^3$, a line $z_1 + z_2 = c$ meeting the hyperbola $z_1 z_2 = m^3$, and step 4 is a cube root chosen so that $uv = m$, which uses the hyperbola to pair the two cube roots correctly.

3 / THE DISCRIMINANT / THREE CASES

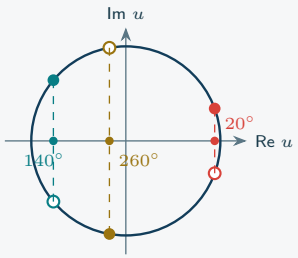
The line $z_1 + z_2 = c$ meets $z_1 z_2 = m^3$ when $c^2 \geq 4m^3$, which is $\frac{q^2}{4} + \frac{p^3}{27} \geq 0$

case	line against hyperbola	the cubic	example
> 0	two real crossings	one real root, two complex	$t^3 + 3t - 4$: $1,\ -\frac{1}{2} \pm \frac{\sqrt{15}}{2}i$
$= 0$	tangent, $u = v$	a repeated root	$t^3 - 3t + 2$: $u = v = -1$, roots $-2, 1, 1$
< 0	no real crossing	three real roots	$t^3 - 3t - 1$: $1.879,\ -0.347,\ -1.532$

The third row is the casus irreducibilis, where Cardano's formula needs the cube root of a complex number although every root is real, and the hyperbola picture shows why: the six crossings of $uv = m$ with $u^3 + v^3 = c$ all leave the real plane.

4 / THREE REAL ROOTS / THE HYPERBOLA BECOMES A CIRCLE

For $m = 1$ the points $(u, v) = (e^{i\theta}, e^{-i\theta})$ give $t = 2\cos\theta$ and $t^3 - 3t = 2\cos 3\theta$



When the line misses the hyperbola in the z-plane the numbers u^3 and v^3 are complex conjugates of modulus $m^{3/2}$, so $u = \sqrt{m}\,e^{i\theta}$ and $v = \sqrt{m}\,e^{-i\theta}$ lie on a circle and $t = u + v = 2\sqrt{m}\cos\theta$ is real. Then $t^3 - 3mt = 2m^{3/2}\cos 3\theta$ is the triple angle formula, so solving $t^3 - 3t = 1$ asks for $\cos 3\theta = \frac{1}{2}$, whose solutions $\theta = 20^\circ,\ 140^\circ,\ 260^\circ$ give the roots

$$t_k = 2\sqrt{m}\cos\frac{\theta_0 + 2\pi k}{3}, \text{ i.e. } 1.879,\ -1.532,\ -0.347$$

The open points are the conjugate partners (v, u), and each root is the common real part of a pair.

5 / THE OTHER ROOTS / ROTATING ALONG THE HYPERBOLA

Multiplying u by ω and v by ω^2 keeps uv fixed, which gives all three roots

There are nine sums $\omega^i u + \omega^j v$, and only the three with $i + j \equiv 0 \pmod 3$ keep $uv = m$, namely $u + v$, $\omega u + \omega^2 v$ and $\omega^2 u + \omega v$. For $t^3 + 3t - 4 = 0$ these are 1 and $-\frac{1}{2} \pm \frac{\sqrt{3}}{2}(u - v)i = -\frac{1}{2} \pm \frac{\sqrt{15}}{2}i$, since $u - v = \sqrt{5}$, and their sum is 0 and their product $4 = -q$, as it must be.

ASIDE / VIETA'S SUBSTITUTION

Setting $t = u + \frac{m}{u}$ is Vieta's version of the trick, and $u^3 + \frac{m^3}{u^3} = t^3 - 3mt$ is a Dickson polynomial.

ASIDE / CHECK IN PYTHON

`np.roots([1,0,3,-4])` gives $1,\ -0.5 \pm 1.936j$, and $\varphi - 1/\varphi = 1$.

Convention. The reduced cubic is $t^3 + pt + q = 0$, $m = -p/3$, $c = -q$, $\omega = e^{2\pi i/3}$. After the earlier sheets on the quadratic as a line meeting a hyperbola.

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